Experimental Methods in Computer Science

(*Metodologias Experimentais em Informática)*

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Correlation

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Correlation

- Correlation is the extent to which two variables are related.
- If the two variables are highly related, then knowing the value of one of them will allow you to predict the other variable with considerable accuracy.
- The less related the variable, the less accurate your ability to predict when you know the other.
- Correlation does not imply causation → do not think in terms of dependent and independent variables

  (causation will result in correlation, but correlation does not necessarily result in causation)

The scatterplot: graphing correlations

- Scatterplots allow us to visually see the relation between two variables.
- One variable is plotted on the ordinate (y) and the other on the abscissa (x).
- **Positive correlations** – occur when both variables move in the same direction (e.g., as the amount of memory increases, the number of transactions per second in a database also increases).
- **Negative correlations** – occur when one variable increases, the other decreases (e.g., as CPU clock frequency increases, the program execution time decreases).

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The scatterplot: graphing correlations

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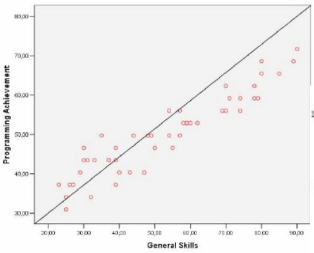

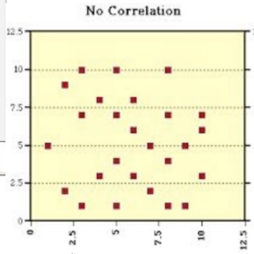

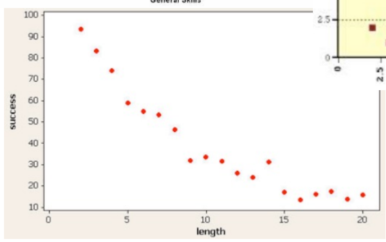

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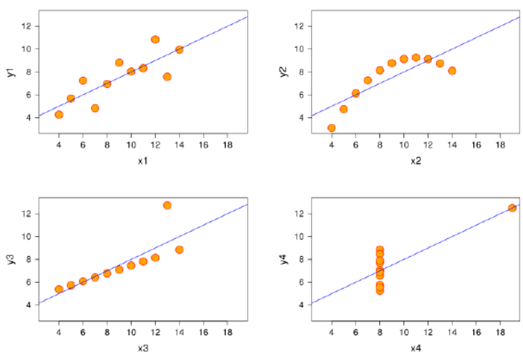
Same parameters; different distributions



The four y variables have the same mean (7.5), standard deviation (4.12), correlation (0.81) and regression line (y= 3 + 0.5x).

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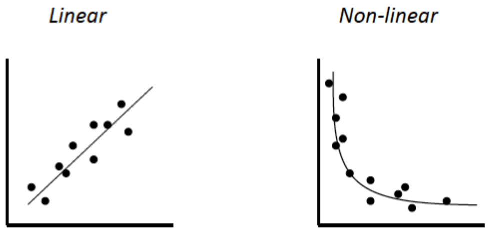
Pearson's correlation coefficient

- Pearson's product-moment correlation coefficient (represented by ***r***) is a measure of the intensity of the *linear* association between variables.
- It is possible to have *non-linear* associations.
- Need to examine data closely to determine if any association exhibits linearity



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Pearson's correlation coefficient (cont.)

- The raw formula for r

$$r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{[n\sum X^2 - (\sum X)^2]\,[n\sum Y^2 - (\sum Y)^2]}}$$

- r is a **unit-less number** between -1 and +1.
  - -1 means full negative correlation; +1 means full positive correlation; 0 means no correlation at all
- If variables are highly correlated:
  - We may want to investigate their association further to determine if there is a causal mechanism operating
  - For independent variables of an experiment, we may use only one to **simplifying the experiment**

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How to calculate *r* (Pearson's coefficient)?

- In R, Excel, SPSS you can calculate *r* directly from samples of X and Y.
- There are many online calculators available
- It is also very easy to calculate *r* manually. Observe the example in the next slides.

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Example: Pearson's coefficient *r* calculation

The following table shows values for the size of a file and the time needed to read the file from a SSD disk in a given computer.

File size in bytes (x_i)	Time in µs (y_i)
10	3.8
50	8.1
100	11.9
500	55.6
1000	97.8
5000	500.2
10000	1006.1
50000	4102.4
100000	9903.4
150000	12782.2

$$r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{[n\sum X^2 - (\sum X)^2]\,[n\sum Y^2 - (\sum Y)^2]}}$$

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Example:
Pearson's coefficient *r* calculation (cont.)

Using a scatterplot we can easily see that there is a clear positive correlation between the two variable.

File size in bytes (x_i)	Time in µs (y_i)
10	3.8
50	8.1
100	11.9
500	55.6
1000	97.8
5000	500.2
10000	1006.1
50000	4102.4
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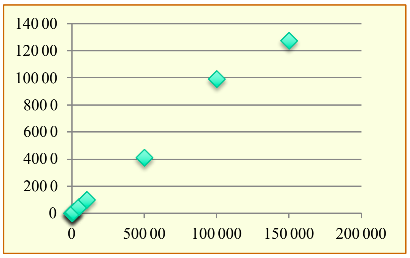

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Example:
Pearson's coefficient *r* calculation (cont.)

Add extra columns to the table to calculate the intermediate terms.

$$r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{[n\sum X^2 - (\sum X)^2]\,[n\sum Y^2 - (\sum Y)^2]}}$$

	File size in bytes (x_i)	Time in ms (y_i)	XY	X^2	Y^2
	10	3.8	38	100	14.44
	50	8.1	405	2500	65.61
	100	11.9	1190	10000	141.61
	500	55.6	27800	250000	3091.36
	1000	97.8	97800	1000000	9564.84
	5000	500.2	2501000	25000000	250200.04
	10000	1006.1	10061000	100000000	1012237.21
	50000	4102.4	205120000	2500000000	16829685.76
	100000	9903.4	990340000	10000000000	98077331.56
	150000	12782.2	1917330000	22500000000	163384636.8
→ $\sum$	316660	28471.5	3125479233	35126262600	279566969.3

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Significance testing for *r*

- If the data are approximately normally distributed we can calculate a **t-statistic** for the correlation coefficient ***r*** using the following formula (assuming n< 30 and using T Student):

$$t = \frac{r}{\sqrt{\frac{1-(r)^2}{n-2}}}$$

- Consider *df* = n-2… since there is one *df* for each column
- In n ≥30 use Z table (standard normal distribution)

Important: with this testing we can test the hypothesis that 2 variables are correlated with a given level of significance (e.g., for α = 0.05)

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Example: Pearson's coefficient *r* calculation (cont.)

Add extra columns to the table to calculate the intermediate terms.

$$r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{[n\sum X^2 - (\sum X)^2]\,[n\sum Y^2 - (\sum Y)^2]}}\ (making\ the\ calculations) = 0.996$$

File size in bytes (x_i)	Time in ms (y_i)	XY	X^2	Y^2
10	3.8	38	100	14.44
50	8.1	405	2500	65.61
100	11.9	1190	10000	141.61
500	55.6	27800	250000	3091.36
1000	97.8	97800	1000000	9564.84
5000	500.2	2501000	25000000	250200.04
10000	1006.1	10061000	100000000	1012237.21
50000	4102.4	205120000	2500000000	16829685.76
100000	9903.4	990340000	10000000000	98077331.56
150000	12782.2	1917330000	22500000000	163384636.8
→ ∑ 316660	28471.5	3125479233	35126262600	279566969.3

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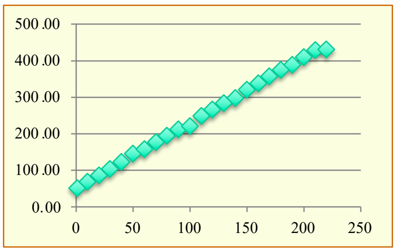
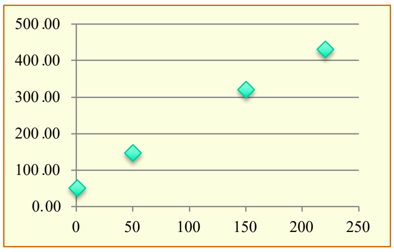
Significance of Pearson's coefficient





The correlation (and the value of r) is similar in the two examples above. But the level significance is quite different.

Important: the significance of the value obtained for r is dependent on the number of samples/points (value of n). As a rule, **use always more than 10**.

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Hypothesis testing for Pearson's correlation coefficient

Follows the same basic steps of the other hypothesis testing:

1. State the hypothesis to be tested
2. Compute the test statistic
3. Obtain p value
4. Make a decision

Pearson's correlation coefficient *r*

$$t = \frac{r}{\sqrt{\frac{1-(r)^2}{n-2}}}$$

Number of samples

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Example of hypothesis testing for Pearson's correlation coefficient

The following table shows values of two variables:

Y – number of lines of code of programs (in thousands KLoC)

X – size of the corresponding files (in Kbytes)

KLoC	Size of file (Kbytes)
0.1	0.07
1	0.80
2	1.65
4	3.35
10	8.40
15	12.50
20	16.50
50	42.00
100	85.00
500	420.50
1000	830.00

Test an hypothesis for whether there is a significant correlation between two variables or not, with a given confidence level (e.g., 95%)

Example: hypothesis testing for Pearson's coefficient (*r*)
Step 1 - State the hypothesis be tested

- **H_0: $r = 0$**
  There is no correlation between the number of lines of code (LoC) and the size of the corresponding file
- **H_1: > 0**
  There is a positive correlation: the larger the LoC, the larger the size of the corresponding file (**Claim)**.

Example: hypothesis testing for Pearson's coefficient (*r*)
Step 2 - Compute the test statistic

Add extra columns to the table to calculate the intermediate terms and facilitate the calculation of *r*.

$$r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{[n\sum X^2 - (\sum X)^2]\,[n\sum Y^2 - (\sum Y)^2]}}$$

KLoC	Size of file (Kbytes)	XY	X^2	Y^2
0.1	0.07	0.007	0.010	0.004
1	0.80	0.8	1.0	0.6
2	1.50	3.0	4.0	2.3
4	2.50	10.0	16.0	6.3
10	5.60	56.0	100.0	31.4
15	8.23	123.5	225.0	67.7
20	12.00	240.0	400.0	144.0
50	32.50	1625.0	2500.0	1056.3
100	85.00	8500.0	10000.0	7225.0
500	98.40	49200.0	250000.0	9682.6
1000	120.00	120000.0	1000000.0	14400.0
1702.1	**366.6**	**179758.3**	**1263246.0**	**32616.0**

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Example: hypothesis testing for Pearson's coefficient (*r*)
Step 2 - Compute the test statistic (cont.)

Add extra columns to the table to calculate the intermediate terms and facilitate the calculation of *r*.

$$r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{[n\sum X^2 - (\sum X)^2]\,[n\sum Y^2 - (\sum Y)^2]}}\ (doing\ the\ calculations) = 0.8615$$

The test statistics

$$t = \frac{r}{\sqrt{\frac{1-(r)^2}{n-2}}} = \frac{0.8615}{\sqrt{\frac{1-(0.8615)^2}{11-2}}} = 5.089$$

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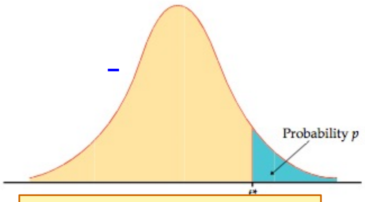
Example: hypothesis testing for Pearson's coefficient (*r*)
Step 3 – Obtain p value

Find the probability for
t = 5.089
n = 11 → df = 11–2 = 9



the critical value t^* with probability p lying to its right and probability C lying between $-t^*$ and t^*.

The probability is smaller than 0.05%

TABLE D
t distribution critical values

	Upper-tail probability *p*											
df	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.00[illegible]	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	[illegible]
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	[illegible]
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	[illegible]
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	[illegible]
5	0.727	0.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	[illegible]
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	[illegible]
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	[illegible]
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	[illegible]
9	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	[illegible]
11	0.697	0.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073

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Example: hypothesis testing for Pearson's coefficient (*r*)
Step 4 – Make a decision

Small ***p*** values provide evidence against the null hypothesis, as it means that the observed data are unlikely when the null hypothesis is true.

Conventions:

- ***p*** **≥ 0.10** → **the observed difference is "not significant"**
- ***0.05 ≤ p < 0.10*** → **the observed difference is "marginally significant"**
- ***0.01 ≤ p < 0.05*** → **the observed difference is "significant"**
- ***p*** **< 0.01** → **the observed difference is "highly significant"**

As ***p*** is smaller than 0.05% the effect is **highly** significant.
We reject H_0 with (more than) 95% of confidence.

We prove that there is a positive correlation between the number of lines of code (LoC) and the size of the corresponding file with 95% of confidence.

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Sample size and significance

- Take into account that ***r*** is influenced by samples size

Sample size, n	Minimum absolute value of r needed to attain significance (using $\alpha = 0.05$)
15	.514
20	.444
30	.361
50	.279
100	.197
250	.124

For large n, r_{crit} is approximately $2/\sqrt{n}$. Table taken from Paul G. Marr slides

- But if the sample size is large, even small r values are important to attain significance

Other ways to measure correlation

- When variables are normally distributed, the **Pearson correlation coefficient** is the appropriate choice.
- For non-normal data or outliers, use **Spearman's rank correlation coefficient**. It is based on based on ranks and captures monotonic relationships.
- For data with many ties or ordinal data, use Kendall's tau. It is is a non-parametric statistic that measures the ordinal association (i.e., the strength and direction of the relationship) between two ranked variables.
- There are many online calculators… For example: https://www.statskingdom.com/correlation-calculator.htmlv